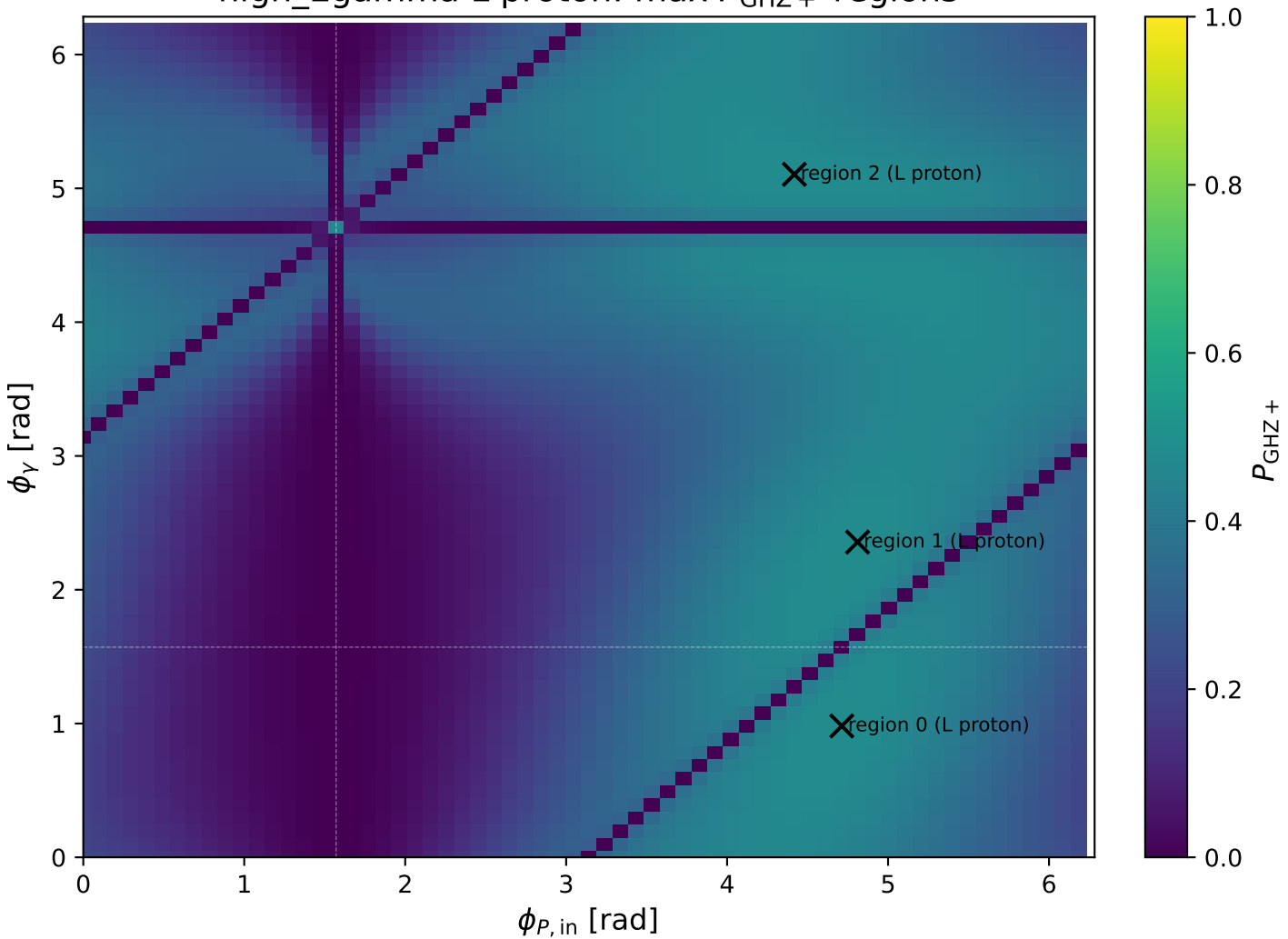
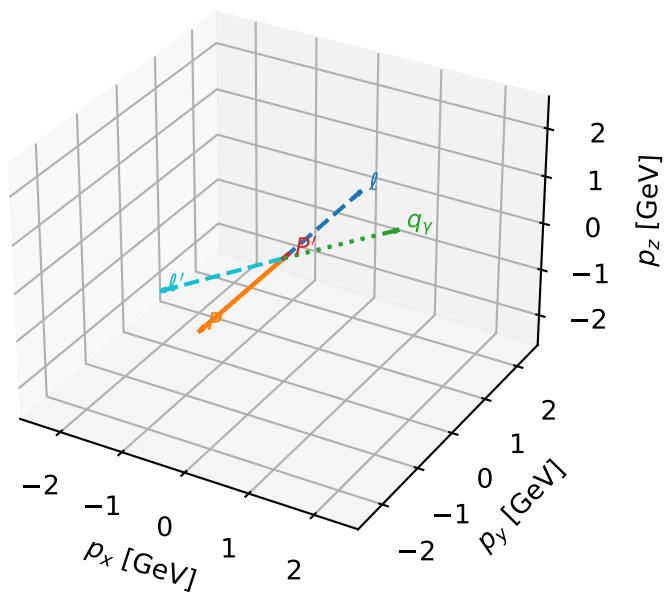


high\_Egamma L proton: max  $P_{\text{GHZ}+}$  regions



# L proton characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

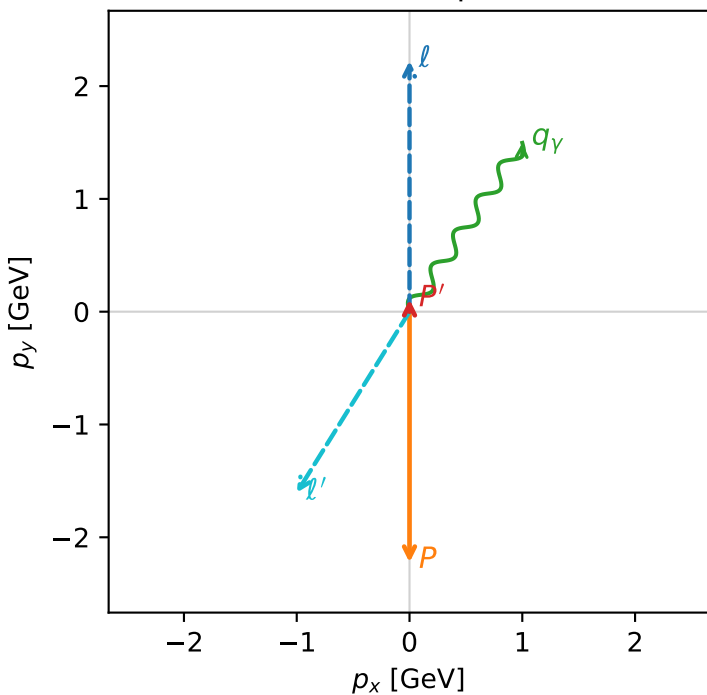


ghz\_purity\_high\_egamma\_l\_proton\_region\_0  $P_{\text{GHZ}}=0.482446$   
region: high\_Egamma L\_proton\_region\_0  
outgoing-state purity: 0.98449  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $\frac{1}{2} \sum_{h_\mu} \rho(h_\mu, L_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=0.108644$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_l=1.5708$ ,  $\phi_P=4.71239$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=0.981748$   
 $Q^2=15.5469$ ,  $x_B=0.834909$ ,  $t=-3.28022$ ,  $W^2=3.95402$ ,  $y=0.902847$   
 $\theta(l', \gamma)=178.171$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= 0.0000$   $p_y= 2.2231$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -0.0000$   $p_y= -2.2231$   $p_z= 0.0000$   
 $l'$   $E= 1.8942$   $p_x= -1.0000$   $p_y= -1.6053$   $p_z= 0.0000$   
 $P'$   $E= 0.9443$   $p_x= 0.0000$   $p_y= 0.1086$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= 1.0000$   $p_y= 1.4966$   $p_z= 0.0000$

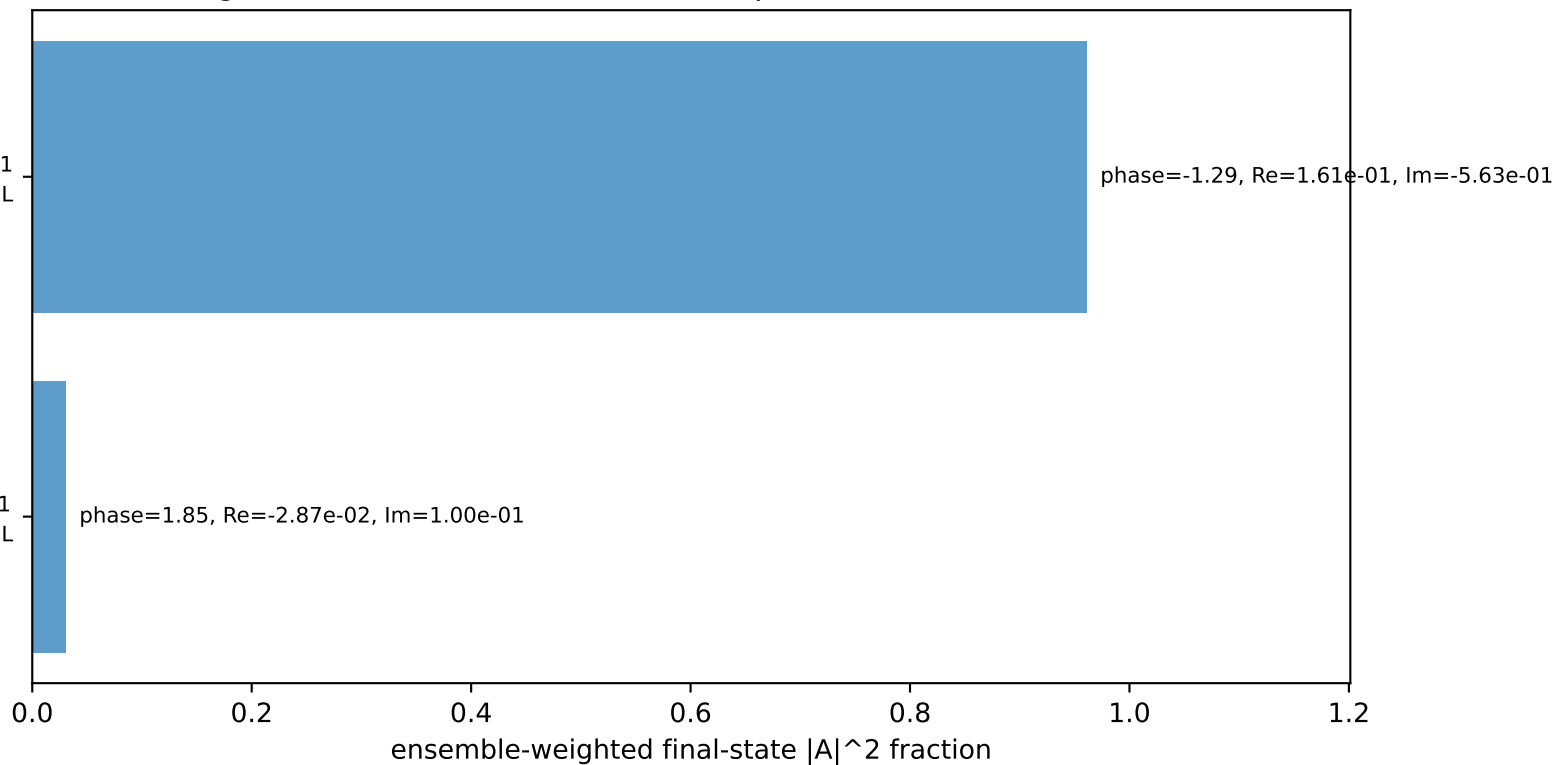
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$   
electron  $h=+1$ , proton L

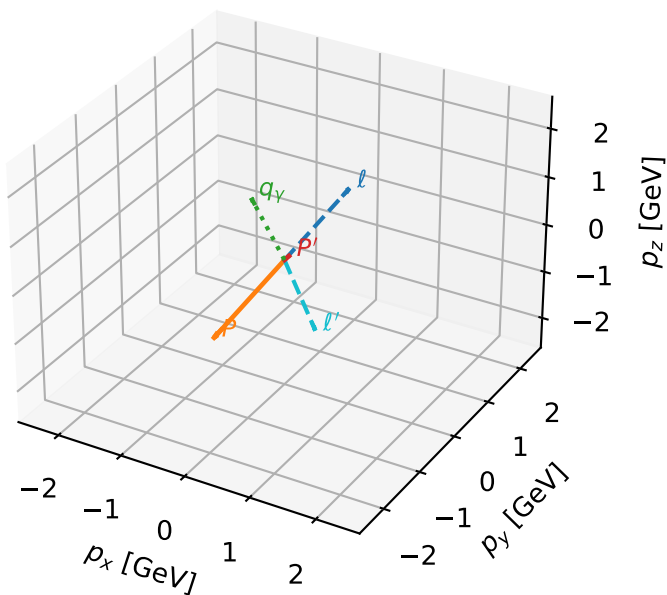
$h_e=+1, h_p=+1, h_\gamma=-1$   
electron  $h=+1$ , proton L

Leading initial-ensemble/final-state components (N=2, retained=99.2%)



# L proton characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

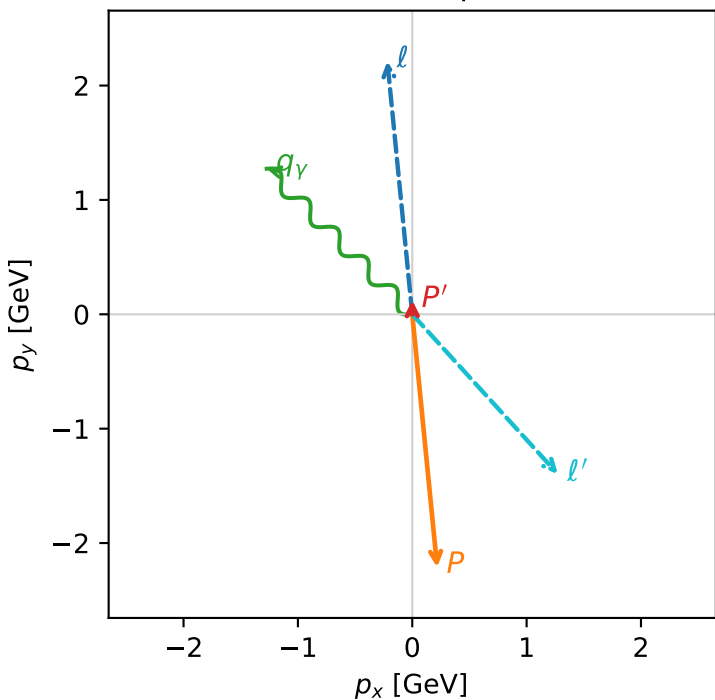


ghz\_purity\_high\_egamma\_l\_proton\_region\_1  $P_{\{\text{rm GHZ}\}}=0.480795$   
region: high\_Egamma\_L\_proton\_region\_1  
outgoing-state purity: 0.98056  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $\frac{1}{2}\sum_{h_\mu}\rho(h_\mu, L_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=0.123642$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_\ell=1.66897$ ,  $\phi_P=4.81056$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=2.35619$   
 $Q^2=15.1349$ ,  $x_B=0.830394$ ,  $t=-3.35315$ ,  $W^2=3.97111$ ,  $y=0.883699$   
 $\theta(\ell', \gamma)=177.348$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2256$   $p_x= -0.2179$   $p_y= 2.2124$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 0.2179$   $p_y= -2.2124$   $p_z= 0.0000$   
 $\ell'$   $E= 1.8924$   $p_x= 1.2728$   $p_y= -1.3964$   $p_z= 0.0000$   
 $P'$   $E= 0.9461$   $p_x= 0.0000$   $p_y= 0.1236$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -1.2728$   $p_y= 1.2728$   $p_z= 0.0000$

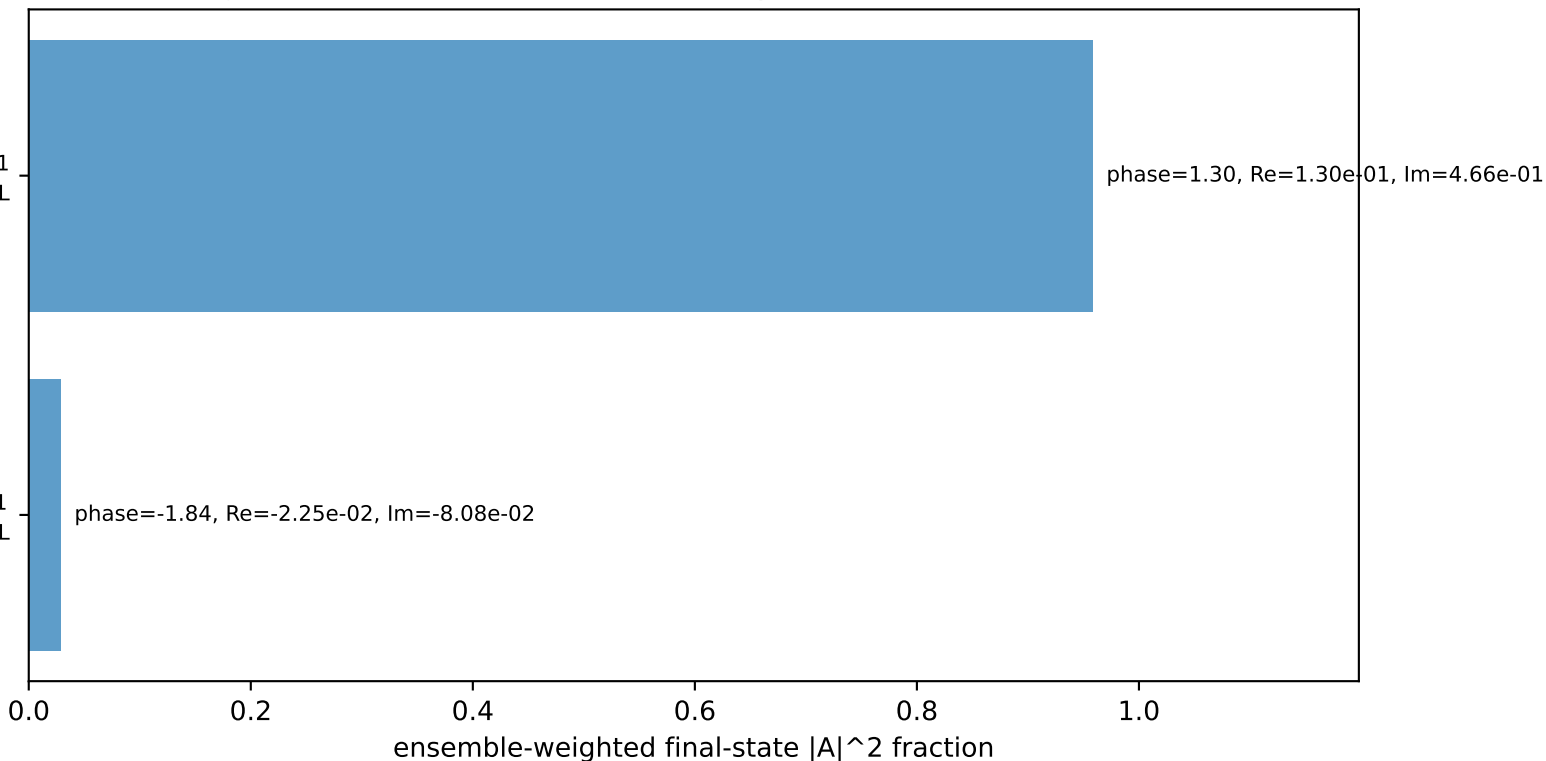
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$   
electron  $h=+1$ , proton L

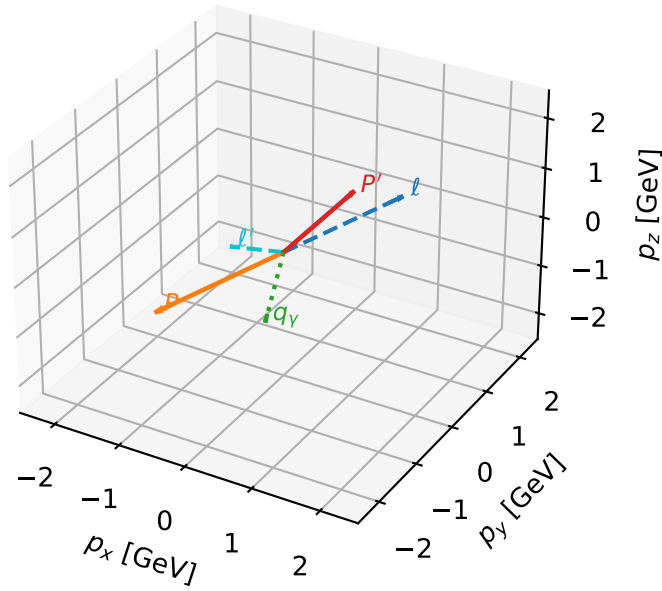
$h_e=+1, h_p=+1, h_\gamma=-1$   
electron  $h=+1$ , proton L

Leading initial-ensemble/final-state components (N=2, retained=98.7%)



# L proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

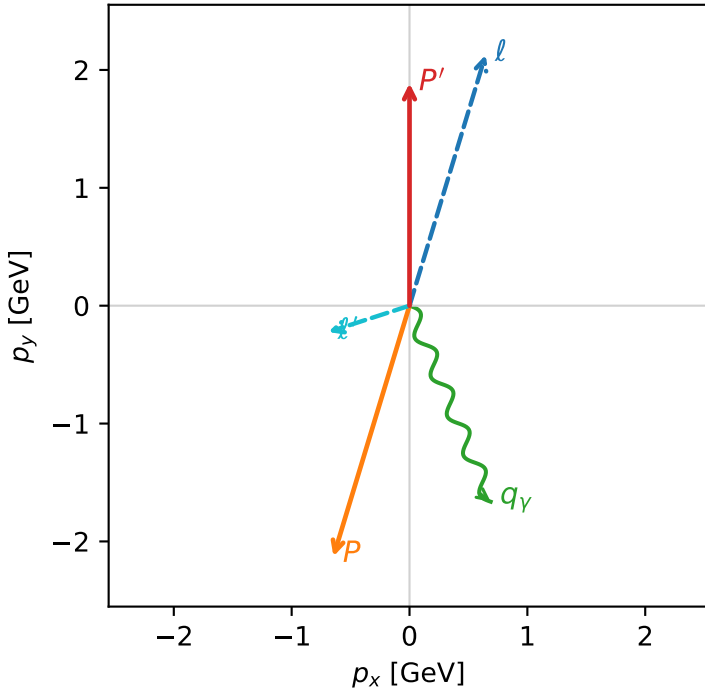


ghz\_purity\_high\_egamma\_l\_proton\_region\_2  $P_{\text{GHZ+}}=0.478826$   
 region: high\_Egamma L\_proton\_region\_2  
 outgoing-state purity: 0.987421  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $\frac{1}{2} \sum_{h_\mu} \rho(h_\mu, L_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=1.88636$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_l=1.27627$ ,  $\phi_p=4.41786$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=5.10509$   
 $Q^2=5.07463$ ,  $x_B=0.268034$ ,  $t=-16.4329$ ,  $W^2=14.738$ ,  $y=0.917958$   
 $\theta(l', \gamma)=94.5328$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= 0.6453$   $p_y= 2.1274$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -0.6453$   $p_y= -2.1274$   $p_z= 0.0000$   
 $l'$   $E= 0.7318$   $p_x= -0.6888$   $p_y= -0.2234$   $p_z= 0.0000$   
 $P'$   $E= 2.1067$   $p_x= 0.0000$   $p_y= 1.8864$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= 0.6888$   $p_y= -1.6630$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=99.0%)

